

## Presentation 5

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What did you do?

- I conducted an analysis of TransferToSafeHaven, the chatbot introduced in the FamilyMenu from August 11 to 29, 2025, on whether the introduction of the chatbot in place of DivorceOrParentingMenu (option 2 of the FamilyMenu) for English speaking callers led to less confusion in the legal menu. 'Less confusion' was assessed via two metrics previously used, redundancies (callers repeating menus, in this case multiple visits to the legal menu) and successful calls (callers reaching a desired destination in the call journey such as a dead end / queue).

How does it help the project?

- The analysis found that around 50% of callers entering the FamilyMenu during their call journey pressed option 2 of the FamilyMenu at some point. Historically without the chatbot, the callers advancing further in the FamilyMenu by choosing option 2 (DivorceOrParentingMenu) revisited the legal menu 50% less than the FamilyMenu callers not choosing option 2 (~11% vs ~22%). Additionally, when comparing callers pressing option 2 on whether they entered the chatbot during Aug 11 – 29 (depending on them being an English or Spanish speaker), there was no difference in legalmenu revisit percentage (both around 13%), suggesting this specific chatbot does not have an effect on reducing the legalmenu revisit rate. This is supported by graphs that display which submenus callers visit after their first and last visit to the legal menu (for familymenu callers visiting the legalmenu multiple times). When filtering for just callers that had chosen option 2, nearly all these callers' last submenu was the familymenu, with only ~3% of callers visiting a different submenu after having visited option 2 of the familymenu. This reveals that the redundancies experienced in these caller's journeys are happening before they press option 2 of the family menu; thus, a chatbot does not help in this regard. Compared with the callers in the familymenu not choosing option 2 at any point, however, for around 27% of them, their final submenu from the legalmenu was not the familymenu, meaning the inclusion of a chatbot directly upon entering the familymenu would help assist with the confusion of these callers.
- Additionally, extending the definition of a successful call to reaching the chatbot, the success rate of calls marginally improves in the familymenu, since callers in the

familymenu have around a 90% success rate (similar across callers choosing and not choosing option 2), so any callers reaching the chatbot will increase this success rate. Thus, if maximizing the success rate of calls is a priority, adding a chatbot to submenus with lower call success rates (such as the housing menu) would bring utility in this regard.

Issues faced (if any)

- No additional since presentation 3

Attempts to resolve issues (if any)

- Refer to presentation 3 documentation

Issues resolved (if any)

- Refer to presentation 3 documentation

Next steps

- Look further into whether having a chatbot upon entering the familymenu or housingmenu would reduce confusion in the legalmenu, or could serve utility in any other measurable regard that can be explored with the data.
- Assess whether introducing a chatbot to the otherlegalmenu is a viable alternative usage for the chatbot
  - o Specifically to collect data on what issues fall under OtherLegalOtherMenu
    - Given the vast majority of callers that choose the otherlegalmenu end up at otherlegalothermenu, meaning their potential legal issues are not directly identifiable by the call journey data, a chatbot that helps identify these caller's issues could also help identify areas of interest that LegalAid can expand their services into, or give LegalAid better guidance on what information they can provide to these callers.

References

N/A

Presentation 4

Link to presentation / code

[Presentation](#)

## Code

What did you do?

There are three main questions I answered for this presentation.

The first is whether we can change the numbering of the legal menu. Specifically, whether this would be feasible given how many people have memorized the keys they have to press to navigate to their desired menu (signaled by spending an extremely short amount of time in the legal menu). For this analysis, I created histograms for time spent during the first legal menu visit separated by submenu, as well as a total plot for all individuals who navigate to the legal menu and then some submenu.

The second is seeing the breakdown of people navigating to the other legal menu (from the legal menu). Notably from the last presentation, it was observed that a decent proportion of callers enter a different submenu (and particularly seen with the family submenu), and then reenter the legal menu to finally end up at the other legal menu. Thus, I generated plots which submenu callers first reach after entering the other legal menu (whether that is OtherLegalCriminalCase, OtherLegalPersonalInjury, OtherLegalOther, a repeating of the OtherLegalMenu, abandoning at the OtherLegalMenu, or going back to the main menu). I presented these plots for different cohorts of callers. First is all callers whose first submenu is OtherLegalMenu. Next was splitting this cohort into callers who revisited the legal menu, and callers who never returned to the legal menu. Finally, I examined callers whose first submenu was not OtherLegalMenu, and then renavigated to the legal menu, and finally ended up at the OtherLegalMenu, with an extra plot for callers of this cohort whose first submenu was the FamilyMenu.

The third is asking where people are abandoning when they navigate to the housing menu. From the last presentation, for callers navigating to the housing menu after visiting the legal menu multiple times, there was a greater proportion of callers abandoning in the housing menu than other similar submenus of the legal menu. I considered two cohorts of callers, those whose first submenu is housing and never return to the legal menu, and those whose final submenu is housing and had previously visited the legal menu multiple times. Then, I had two sets of plots. The first is the maximum depth callers reached in the housing submenu (whether they reached a queue / voicemail, TenantDeterrence, Tenant, PreTenant, or Housing). The next is where callers eventually ended up / abandoned (not included in the presentation, but within the code).

How does it help the project?

The analysis for the first question showed that a significantly low amount of callers choose a key within one second of being in the legal menu (only 0.1%), and only 3.1% of callers

choose a key within nine seconds of being in the legal menu (when the prompt for HIV / key 1 is read). This suggests a rather low amount of callers are memorizing the keys of the legal menu. Without any other preference for having HIV menu as the first submenu, given that there are less callers navigating to the HIV menu than Housing, Family, Benefits, Criminal Records, Voicemail, Immigration, and Employment, and from prior presentations, the abnormally large amount of call abandonments in the HIV menu suggests mostly impatient callers pressing 1, that the HIV submenu option can be moved to a later key option. With Housing as the most visited submenu overall, and as the submenu with the most visits after a caller first presses 1 and then renavigates to the legal menu, then having Housing as key 1 will absorb many of these impatient callers. Ultimately, the tentative recommendation is to move the HIV submenu to option 8 (right after Employment), and move the existing options 2-8 to 1-7, thus saving about ten seconds of time spent in the legal menu for the vast majority of callers not navigating to other legal.

The second analysis showed that the vast majority of callers reaching the other legal menu do not have an issue that is currently delineated by LegalAid (i.e. about 70% navigate to OtherLegalOtherMenu). Thus, the data will not show what issues callers may have that caused them to initially choose a different submenu and then navigate to the other legal menu (for example, what family-adjacent issues callers have that cause them to navigate first to the family submenu, and then the other legal menu). However, this breakdown is still a descriptive analysis that motivates Vince's analysis of potentially restructuring the other legal menu.

The third analysis shows that for the ~30% of callers ending up in the housing submenu that do not reach a queue / voicemail, about ~15% of the callers reach the TenantDeterrenceMenu, ~7% reach the TenantMenu, and about 3% reach the PreTenantMenu or the HousingMenu, respectively. Notably, the verbiage of the TenantMenu has three options for queues, and then the last option states, 'if none of the prior options apply to you, press 4'. This submenu should have an option for every caller that reaches the TenantMenu, whether that is entering a queue, or the TenantDeterrenceMenu for anyone who does not enter a queue as their issue is not serviceable by LegalAid. This suggests that the ~7% of callers in the housing submenu that only reach TenantMenu would be better served by hearing the affiliate information in the TenantDeterrenceMenu. Thus, the verbiage for option 4 should more explicitly state this option has useful information for these callers.

Issues faced (if any)

- No additional since presentation 3

Attempts to resolve issues (if any)

- Refer to presentation 3 documentation

Issues resolved (if any)

- Refer to presentation 3 documentation

Next steps

Confirming which specific option HIV should be moved in the legal menu

- Current recommendation is option 8, but based mostly on overall traffic in the legal menu

Finalizing option order of legal menu based on total traffic / behavior of repeat visits to legal menu

Regarding the recommendation to update the verbiage of TenantMenu option 4, when exactly do the callers abandon / leave the TenantMenu? Are they leaving before or after having heard option 4?

- If most hear option 4, then this would suggest that, to a lot of these callers, the prompt currently signals a lack of purpose / useful information in continuing to the TenantDeterrenceMenu (basically confirming that the TenantMenu option 4 verbiage should signal that this submenu has potentially useful information).

References

N/A

Presentation 3

Link to presentation / code

[Presentation](#)

[Code](#)

What did you do?

The main part of my analysis involved understanding which submenus callers eventually end up when they revisit the legal menu at least once, and also identifying moments with lower call success rates.

First, for the legal menu analysis, I slightly updated the definition of a successful call to include dead end prompts in addition to queues, while keeping percent of callers reaching

a queue as a separate variable. This is especially important for analysis on successful calls for individuals who reach the other legal menu. Without this change, the perception is that a minority of these callers are reaching an intended destination, but in reality, these callers are merely reaching the dead end prompts that direct them to help outside of Legal Aid's services.

In terms of the legal menu analysis, I updated the tree chart giving the proportions of callers who reach the legal menu who listen to the full prompt (or not) and went to the right menu (or not), by including the percent who reach a queue or dead end for each of the four categories of callers reaching the legal menu.

Then, as a continuation of presentation 2, I looked into the subsequent submenus that callers visit after returning to the legal menu from a notable submenu (one with a higher rate of callers exiting out of the submenu back into the legal menu), including HIV, family, housing, other legal, and a direct repeating of the legal menu prompt. For each of these first submenus visited, I displayed the count of callers returning to the legal menu by each submenu callers ended up in after their final visit to the legal menu. I included a second graph displaying the percent of successful calls (visiting a queue or dead end) and the percent of callers reaching a queue (again for each submenu reached after the final visit to the legal menu).

How does it help the project?

This helps with the project in a couple of ways. First, incorporating the 'successful call' variable adds a layer of depth to whether callers are ultimately reaching a desired destination, and whether prompt repetitions or a revisit to the legal menu is impactful and helpful for callers to navigate to the desired queue (or referral to a non Legal Aid service). Additionally, given the early options in the Legal Menu (HIV, family, housing) had disproportionate amounts of callers reverting back to the legal menu, this is important for a greater understanding of whether these callers are confusing two submenus, or choosing an option too early, or where callers have the most confusion, for whether submenu prompts should be reworded or reordered for example. (I will expand upon the notable results with the 'next steps' part.

Issues faced (if any)

1. Discrepancy between the CAR data (from at least January 2025) and the flow chart. Specifically, flowchart implies routing from the main number call menu to the main menu, while CAR data implies it routes instead to the pre-legal seniors menu.

2. Additionally, an additional question surrounding how to identify callers choosing English versus Spanish at the LanguageSelectionMenu (given the English and Spanish versions of the various menu prompts have different lengths)

Attempts to resolve issues (if any)

1. Provided example contact session ID containing issue to Krish
2. To the complex EDA team's knowledge, able to directly identify the language selected by caller from the CAR data. Our compromise was to choose a length of time close to the full English and Spanish prompt length (for example, we used 120 seconds for the LegalMenu2 when the complete prompt length was 129 seconds for English and 157 seconds for Spanish). We determined that this was an appropriate compromise for the sake of analysis, given the callers with a call length between 120-129 seconds (for English speakers) and 120-157 seconds (for Spanish speakers) have effectively listened to the entire prompt (especially if the caller eventually chooses an earlier option, their intention is clear to listen to most/all of the options).

Issues resolved (if any)

Waiting to resolve issues

Next steps

- There is a significant difference in percent of successful calls for those returning to legal menu among those listening to full prompt (80%) vs not listening to full prompt (63%)
  - Next step: understand why
  - Potential reason: large group chooses housing submenu and then reenters (combined with lower call success rate)
- Lowest success rate for ending up at HIV menu
  - Mostly attributable to impatient callers? (as #1 option)
  - Next step: How much time are these callers spending in the HIV menu, as well as the previous submenus? Shorter times may indicate these are impatient callers, but longer times may indicate confusion across submenus.
- Almost  $\frac{1}{3}$  callers swap from family to other legal menu
  - Next step: verbiage analysis of family and other legal menu

- What family-adjacent issues are not directly serviceable in family submenu?
- Potential issues within the housing submenu itself? (Lower success rate of around 65%, alongside a large group of individuals reentering housing menu several times)
  - Next step: housing submenu verbiage analysis

## References

N/A

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## Presentation 2

Link to presentation / code

[Presentation](#)

[Code](#)

What did you do?

I did an analysis on repeated processes in the legal menu. First, I categorized callers who reach and also advance beyond the legal menu into four categories:

1. Listens to all the options and goes to the right menu
2. Listens to all the options, and goes to the wrong menu and goes back
  1. Indicating issue with verbiage
3. Makes a choice before listening to all the options, and goes to the right menu
4. Makes a choice before listening to all the options, and goes to the wrong menu and goes back
  1. Indicating issue with the order in which the option is being presented and / or verbiage

Additionally, I analyzed the frequency of the number of visits callers make to the legal menu, as well as a breakdown of callers making multiple visits to the legal menu (repeating the prompt, or entering a submenu and exiting back to the legal menu). Furthermore, I

determined the percent of calls reaching the legal menu that were ultimately successful, defined as eventually reaching a queue.

The main part of the analysis involved identifying the percent of callers entering the wrong submenu from their first time in the legal menu (meaning they went back to the main menu) by each sub menu. I broke each of the submenu options down by whether the caller listened to the full prompt before making their choice. Finally, I showed the percent of callers listening to the full prompt before choosing a submenu (broken down by submenu). Altogether, this provides the percentages of the four types of callers for each submenu chosen.

How does it help the project?

This helps the project in identifying which submenu options have greater sources of confusion, and also with what specific issues these options may have (whether they contain verbiage issues / issue in ordering / confusion with a specific different menu / etc.). The analysis indicates the early options (HIV, family, housing) have the greatest rates of reaching the wrong menu (meaning they must go back to the legal menu), but the vast majority of callers choosing these submenus do not listen to the full prompt. This indicates callers may be confused a bit with the verbiage, and may also be choosing an option too early (issue with ordering and confusion/overlap with a later submenu option).

Issues faced (if any)

3. Discrepancy between the CAR data (from at least January 2025) and the flow chart. Specifically, flowchart implies routing from the main number call menu to the main menu, while CAR data implies it routes instead to the pre-legal seniors menu.
4. Additionally, an additional question surrounding how to identify callers choosing English versus Spanish at the LanguageSelectionMenu (given the English and Spanish versions of the various menu prompts have different lengths)

Attempts to resolve issues (if any)

3. Provided example contact session ID containing issue to Krish
4. To the complex EDA team's knowledge, able to directly identify the language selected by caller from the CAR data. Our compromise was to choose a length of time close to the full English and Spanish prompt length (for example, we used 120 seconds for the LegalMenu2 when the complete prompt length was 129 seconds for English and 157 seconds for Spanish). We determined that this was an appropriate compromise for the sake of analysis, given the callers with a call length between 120-129 seconds (for English speakers) and 120-157 seconds (for Spanish speakers)

have effectively listened to the entire prompt (especially if the caller eventually chooses an earlier option, their intention is clear to listen to most/all of the options).

Issues resolved (if any)

Waiting to resolve issues

Next steps

- Early options (HIV, Family, Housing) have disproportionate number of revisits to legal menu, despite most callers not listening to full set of options (likely verbiage issue and also an issue with ordering / confusion with a later submenu option)
  - **Next goal #1:** Analyze the submenus chosen on second / third visit to legal menu
    - Do callers confuse two sub-menus?
    - Do callers choose an option too early?
    - Which submenus do callers choose after directly repeating the legal menu prompt?
- In addition to '% right submenu' and '% listening to full prompt' as variables, introducing third variable ('% eventually reaching queue', i.e. successful call journey)
  - **Next Goal #2:** Analyze impact of multiple visits to legal menu on eventual call success
    - High percentage reaching queue after second legal menu prompt (especially for early options), may indicate the potential effectiveness of verbiage improvement
      - I.e. whether the callers initially reaching wrong sub-menu can be matched to their correct sub-menu (or the caller's issue does not apply to any sub-menu / legal aid cannot help)

References

N/A

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Presentation 1

Link to presentation

[Link](#)

What did you do?

Our group (Will, Vince, YiHyun) collectively defined redundancies of the call journeys and partitioned them into three types: calls taking longer than they should, repeated processes, and coding/internal redundancies. For each redundancy, we identified different subclassifications of the given redundancy, and the outline for an algorithm to systematize the locating of these redundancies. I focused mostly on the third redundancy (coding / internal redundancies), identifying a redundant checking of the main hours in the main call menu when a caller does not choose a language (and is prompted to the front desk rather than the main menu). Additionally, I helped with the brainstorming of the different subtypes of repeated process redundancies (type 2) and their respective algorithms.

How does it help the project?

This brainstorming of defining/classifying different categories of redundancies serves as a solid framework, where we can utilize our algorithms to more systematically identify particular locations in the call journey that provide either an unnecessary long or confusing experience for the caller.

Issues faced (if any)

Discrepancy between the CAR data (from at least January 2025) and the flow chart. Specifically, flowchart implies routing from the main number call menu to the main menu, while CAR data implies it routes instead to the pre-legal seniors menu.

Attempts to resolve issues (if any)

Provided example contact session ID containing issue to Krish

Issues resolved (if any)

Waiting to resolve issue

Next steps

Type 3 redundancy:

- Continue flowchart analysis and identify moments where certain business hours can be consolidated, and moments where holidays are checked for a second time.

Type 2 redundancy:

- Focus on each of the four subtypes previously defined (visit to the same node multiple times, move to the previous menu -> return to the same node, move to the previous menu -> return to different node, cycles). For each, implement/code our algorithm framework/intuitions to identify the nodes with the greatest occurrence of each type 2 redundancy subtype (as well as the greatest rate of redundancy, specifically the percent of call journeys reaching a given node that are experiencing this particular redundancy subtype).

#### Type 1 redundancy:

- Same analysis as described in type 2; however, I believe Vince will be mainly looking at these type 1 redundancies (where calls are taking longer than they should)

#### References

- N/A